

SYSTEMS THINKING

Systems Analysis & Design

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Introduction to Systems Thinking
- 2 Systems Properties
- 3 Systems Classification



Outline

1 Introduction to Systems Thinking

2 Systems Properties

3 Systems Classification

2017

Hugging Face

Attention

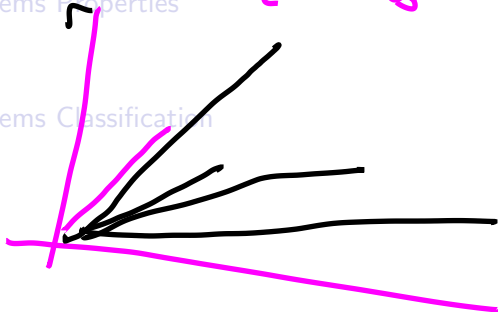
Vector Database

$$w_1 \begin{bmatrix} \theta_1 \\ r=0.8 \end{bmatrix}$$

$$w_2 \begin{bmatrix} \theta_2 \\ r=0.1 \end{bmatrix}$$

$$w_3 \begin{bmatrix} \theta_3 \\ r=0.9 \end{bmatrix}$$

$$w_4 : \begin{bmatrix} \theta \\ \text{Logo} \end{bmatrix}$$



θ



Introduction to Systems Thinking I

- A **system** is a set of interconnected elements with a common purpose.

- Not all elements need to be connected to each other, but every **connection** should be meaningful.

- For some connections there are, the number of connections in a system must be feasible. Representation must be feasible.

- Each element must have at least one connection. Isolated elements make no sense in a system.

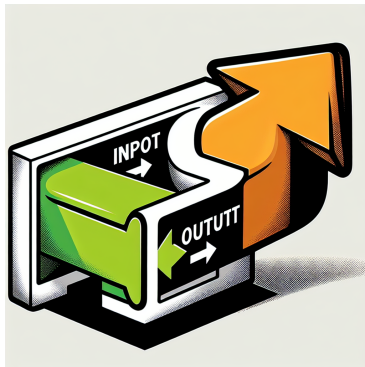


Figure: Prompt: Draw an image of a box with input and output arrows.



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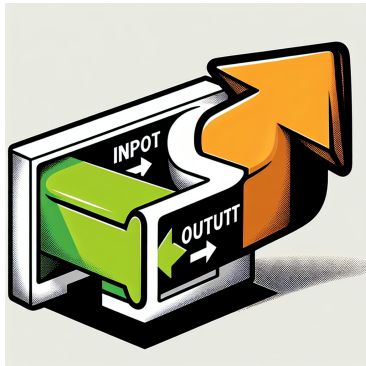


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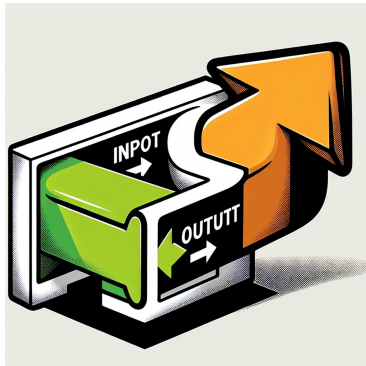


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clients



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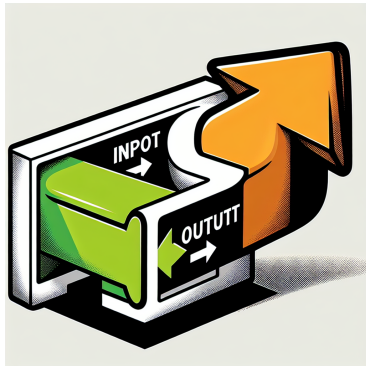
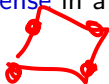
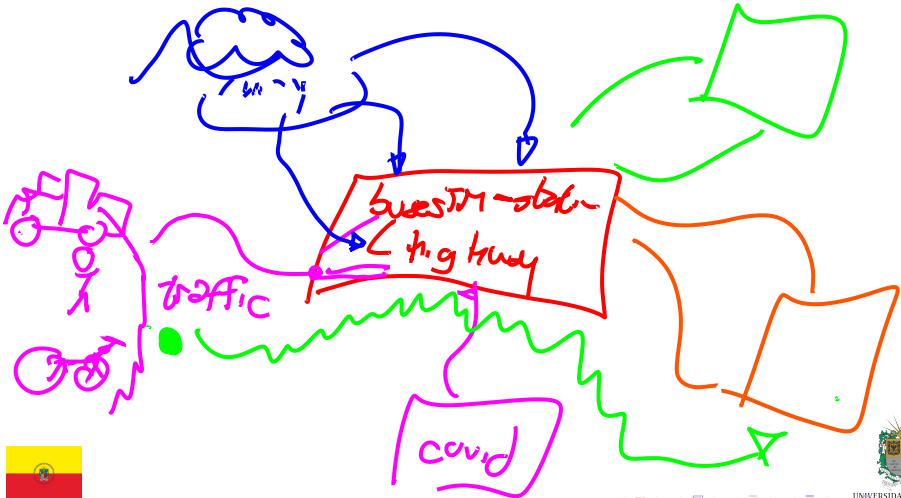


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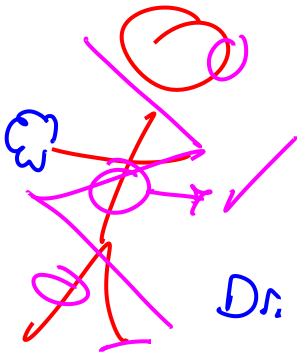


Complexity in Systems

System complexity could be defined as the number of elements and connections in a system.



Introduction to Systems Thinking II



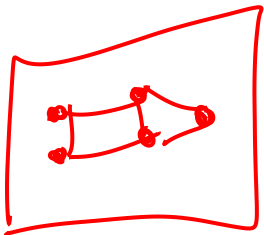
Dr. House

PHP → scalability?

- In **systems thinking**, if you just **split** parts and forget relationships, you will **lost the full picture**.
- It is called **holistic** approach, try to see **all the picture** with all the **meaning details**.
- Define the **box boundaries** is sometimes tricky, as we said, not too complex, not too simple. It is like the desired **universe balance of Thanos**.



Introduction to Systems Thinking II



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Introduction to Systems Thinking III

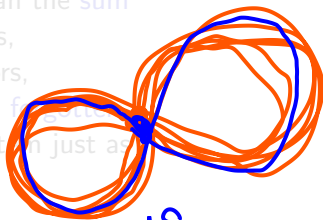
Header (The back)

- Another important concept is the **homeostasis**. It means to put a system in an **equilibrium** state. That is hard, **systems** are both **not in equilibrium** and **resilient to change**. **Chaotic attractors** study is useful here.

- A system is more than the sum of the parts. It means, relationships, behaviors, recovery capacity, are for better or for worse when you see the system just as a sum of its parts.

Lorenz

Machine Learning



S-20
Lines

Data

Dynamical

rule.

+ addresses



Introduction to Systems Thinking III

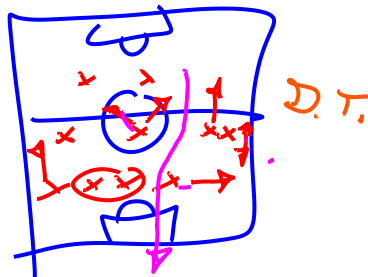
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Introduction to Systems Thinking IV

- Systems thinking** is a way to understand and represent problems in order to find the best possible solution.
- Think in a **problem as a system** lets you understand details, involved elements, relevant information.
- Systems should be viable, auto-sostenible, provides internal feedback loops, and also looks like a whole live-entity.



Introduction to Systems Thinking IV

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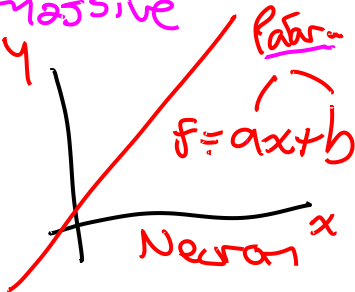
Introduction to Systems Thinking V

calculate mechanical/machine



massive

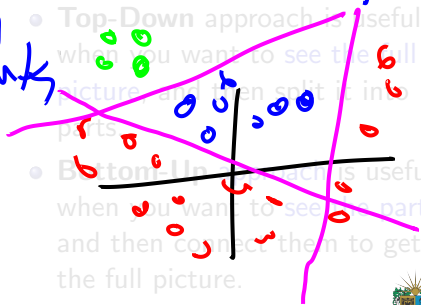
y



dank

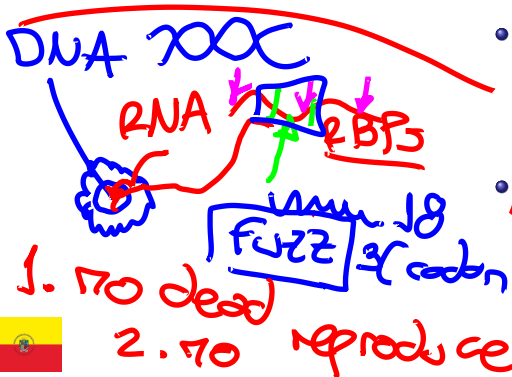
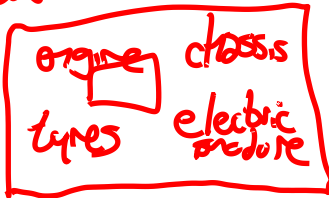
- **Computation** helps to represent behaviors in a mathematical way. Also, it lets to find patterns and information, simplify process; an example of all this is the Artificial Intelligence.

- Top-Down approach is useful when you want to see the full picture, and then split it into parts.
- Bottom-Up approach is useful when you want to see the parts and then connect them to get the full picture.



Introduction to Systems Thinking V

car

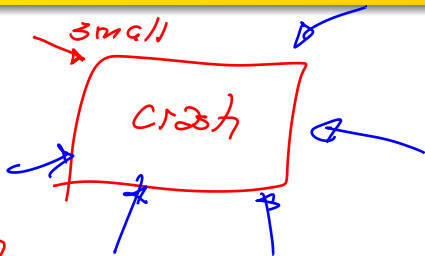


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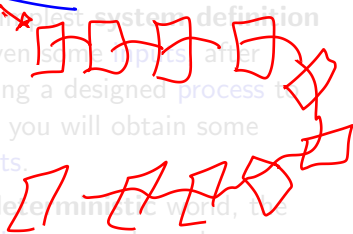


Introduction to Systems Thinking VI

- It is important to understand the **sensitivity** of the problem because it leads to making better decisions.



- The simplest system definition is: given some inputs, after applying a designed process to them, you will obtain some outputs.



- In a deterministic world, the same inputs produce the same outputs. However, real life is not deterministic.



Introduction to Systems Thinking VI

- It is important to understand the **sensitivity** of the problem because it leads to making better decisions.
- The simplest **system definition** is: given some **inputs**, after applying a designed **process** to them, you will obtain some **outputs**.
- In a **deterministic** world, the same **inputs** produce the same **outputs**. However, real life is **not deterministic**.

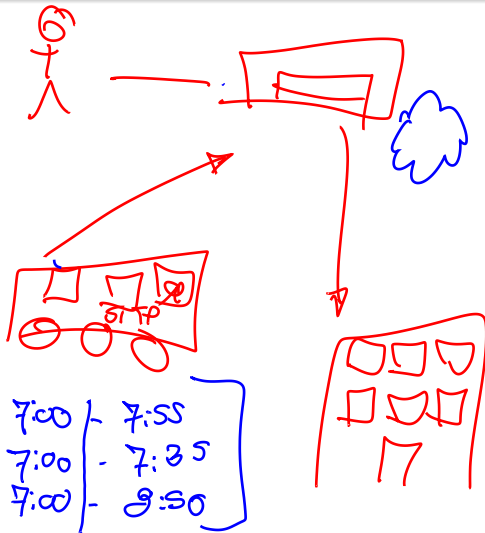


controlled conditions

$$F = m \cdot A.$$



Introduction to Systems Thinking VII



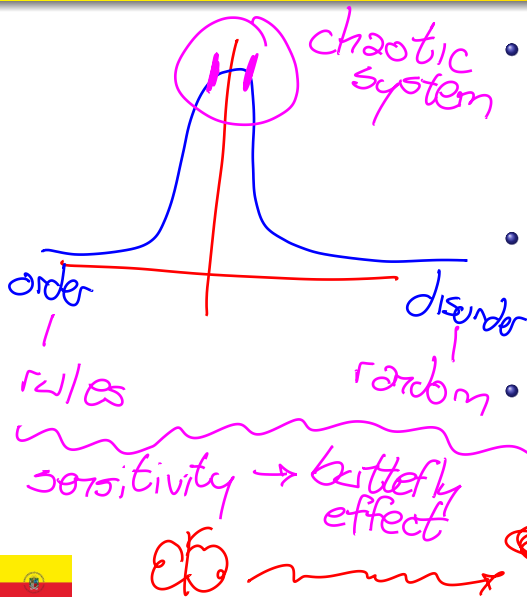
- Since **randomness** is normal in the real world, relying solely on deterministic processes is **dangerous**. Using **stochastic processes** is a better approach.

- Stochastic processes** make use of **probability**, which leads to a better representation of real-world behavior.

- Here **Chaos Theory** becomes a useful tool. To put it simply, chaos can be defined as a harmonious balance between rules and randomness.



Introduction to Systems Thinking VII

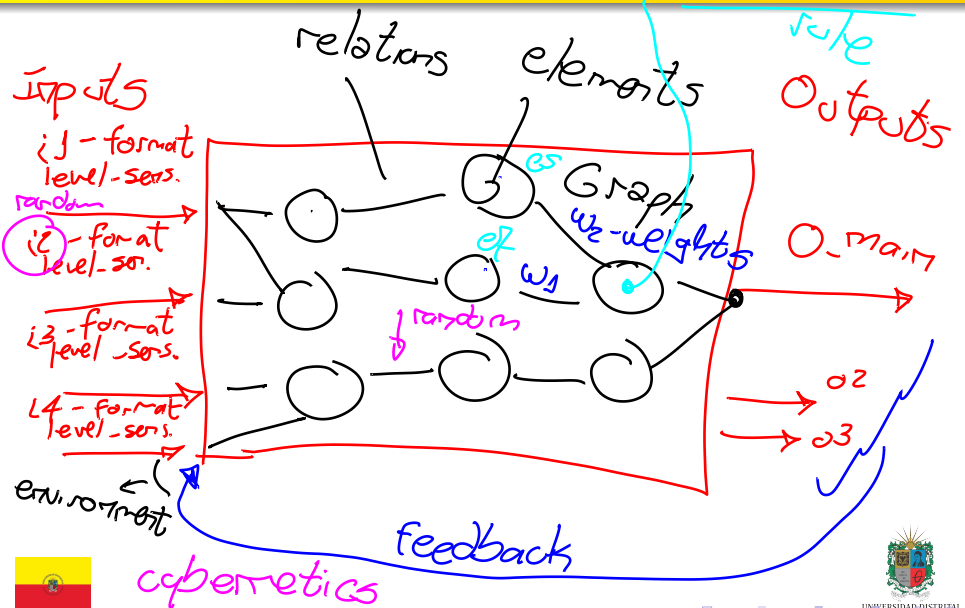


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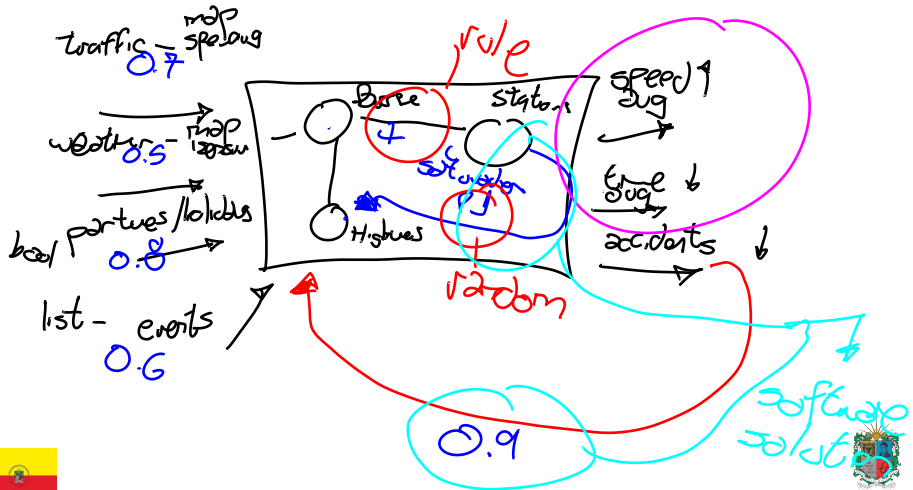


Systems Structure Draw

$e5, w2 + e7, w1$



Case of Study: Transportation System



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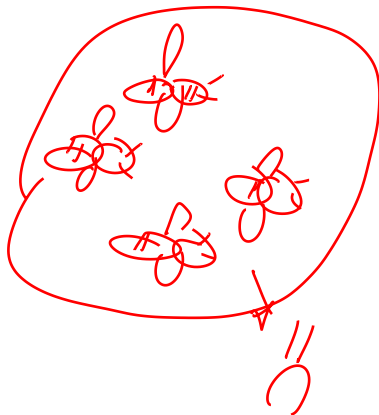


Systems Properties I

- **Emergence** is a property of systems that means that the whole system is more than the sum of its parts.

↙
synergy

- **Interconnectedness** is a property of systems that means that all the elements are connected in a meaningful way.
- **Feedback** is a property of systems that means that the system has internal loops that control the system behavior.
- **Hierarchy** is a property of systems that means that the system has levels of organization.



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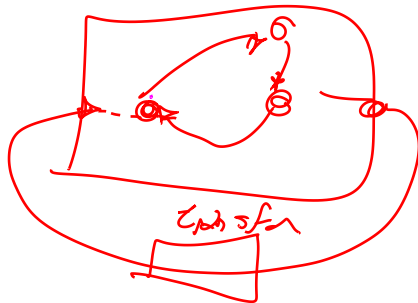
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isolated elements



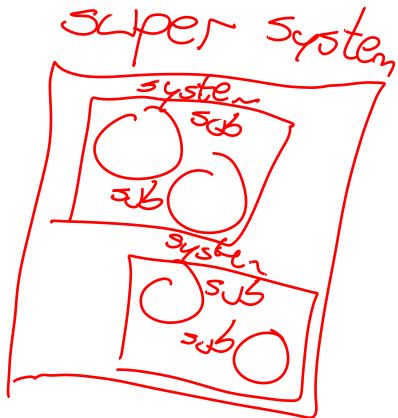
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- **Dissipative** is a property of systems that means that the system can **lose energy** and **information** to the environment.
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Basic Concepts

A **system** can be **classified** according to different **criteria** like **openness**, **adaptability**, **determinism**, and **linearity**.



Systems Classification I

- **Open systems** are systems that can **interact** with the **environment**.
- **Closed systems** are systems that **cannot interact** with the **environment**.
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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

